

USBSID-Player manual

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Disclaimer

I do this stuff in my free time for my enjoyment. Since I like to share my joy in creating this with everyone I try my best to provide a working PCB and Firmware. I am in no way an electronics engineer and can give **no guarantees** that this stuff does not break or damage your hardware, computer, phone, or whatever you try to hook it up to. Be sure to take great care when inserting any real MOS SID chips into the board. While everything has been tested with real chips, this is in no way a guarantee that nothing could go wrong. Use of this board and firmware at your own risk! I am in no way responsible for your damaged hardware. That being said, have fun!

What!?

Yes you read that right, USBSID-Player is embedded on USBSID-Pico and allows you to play PSID, RSID, PRG or P00 chiptunes without the need of an emulator on your computer!

Firmwares with embedded USBSID-Player

Starting with firmware version v0.6.0-BETA firmwares have an embedded emulator that can play PSID and RSID tunes and PSID64 PRG/P00 files.

From v0.8.1 and later these files are for example [usbsidpico-v1.0_em.uf2](#), [usbsidpico2-v1.3-rgb_em.uf2](#) or [usbsidpico2-v1.5_em.uf2](#)

This USBSID-Player is in active development and some tunes might not work or play at all, let me know if you come across any!

NOTE! Due to the hardware limitations of the RP2040, highly demanding, cycle-exact SID tunes (dense per-cycle register writes, e.g. some digi/sample tunes) may not play at full speed on Pico1 or Pico1_w boards. Use a Pico2 or Pico2_w (rp2350) board for these tunes.

WARNING! Whenever the USBSID-Player is compiled into the firmware, the Pico MCU is overclocked to 250MHz, regardless of Pico type. Passive cooling *might/could* be needed.

Cynthcart also runs through the embedded USBSID-Player - see the [USBSID-Pico-Cynthcart-manual](#) file for more information.

Usage

To play SID tunes and run PRG/P00 PSID64 files *on* USBSID-Pico you will need the [send_sid](#) tool that is available in the [tools.zip](#) starting with firmware release v0.6.0-BETA and up, the help menu is self explanatory.

Firmware release v0.7.0 and up supports the send sid mode on the USBSID-Pico config website and on Deepsid!

Multi-subtune PSID/RSID files are supported - you can switch to the next or previous subtune while a tune is playing.

Drivers

Linux Udev rules

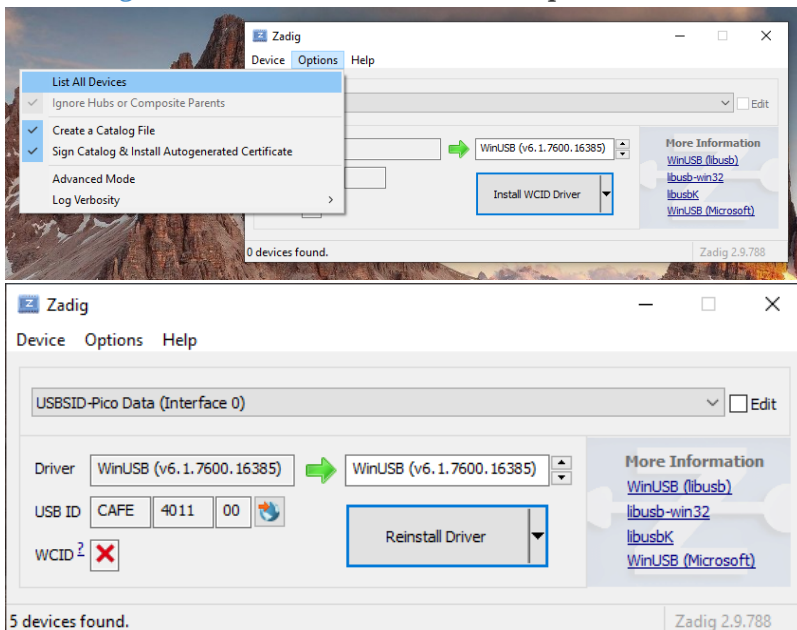
In the [examples/udev](#) directory you can find the [udev rules](#) that I use on Linux.

This purely an example file that you can use and change to your own needs. Steps required for this to work

```
# Check if you are in the plugdev group with the groups command
groups
# If not, add yourself to the plugdev group, then log out and in
sudo usermod -aG plugdev $USER
# Copy the udev rules file to the correct directory
sudo cp 69-usbhid-permissions.rules /etc/udev/rules.d
# Now reload the udev rules
udevadm control --reload-rules && udevadm trigger
# Not working? Try reloading the service
sudo systemctl restart udev
```

Windows driver

Use [Zadig](#) to install the correct driver. Replace the driver for **USBSID-Pico Data** with WinUSB.



Development

More information about USBSID-Player is available on the [github repo](#)

License

Software License ~ GNUv2

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Configtool Software License ~ EPLv2

This program and the accompanying materials are made available under the terms of the Eclipse Public License 2.0 which is available at eclipse.org.

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